AADITYA DESAI

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Computer Engineering undergraduate building AI infrastructure, on-device ML, and embedded software; seeking Summer 2027 software engineering internships.

EDUCATION

San Jose State University | B.S. Computer Engineering — **Expected Spring 2028**
San Jose, CA | GPA: 3.7

De Anza College | Certificate of Achievement, Programming in C/C++ — **Mar 2026**
Cupertino, CA | GPA: 3.8 | Dean's List

Certification: AWS Certified Cloud Practitioner (CLF-C02), Jan 2026

TECHNICAL SKILLS

Languages: C++, C, Python, Kotlin, TypeScript, JavaScript, x86 Assembly (MASM)
ML and Computer Vision: PyTorch, YOLO, OpenCV, LiteRT/TFLite, ML Kit, Hugging Face Transformers
Software, Systems, and Hardware: Linux, Git/GitHub, FastAPI, WebSockets, SvelteKit, Jetpack Compose, Raspberry Pi 5, Qualcomm Hexagon NPU

EXPERIENCE

Technical Lead | VisionAssist, Special Projects — **Spring 2026**
De Anza College, in collaboration with Infineon Technologies | Cupertino, CA

- Led a 5-person team building a head-mounted assistive vision device on Raspberry Pi 5; owned the Python perception pipeline, hardware integration, and iterative testing.
- Built an IMX708 camera-to-inference pipeline with YOLOv8n CPU inference at about 200 ms per frame; ranked detections by risk and proximity for on-device speech alerts.
- Evaluated Infineon 60 GHz radar-camera fusion, diagnosed reflector mismatch, documented findings for Infineon, and replaced radar ranging with calibrated monocular distance estimation.

Teaching Assistant | ENGR 10, Introduction to Engineering — **Aug 2026 - Present**
San Jose State University | San Jose, CA

- Lead two weekly lab sections, totaling 6 hours per week, for an introductory engineering course.

Teaching Assistant and CIS Tutor — **Jan 2025 - Dec 2025**
De Anza College | Cupertino, CA

- Taught C++, x86 assembly, Python, data structures, and OOP; diagnosed algorithmic and memory errors while guiding first-time programmers toward independent debugging.

Co-Founder and Co-President | Applied Data Analytics Club | 44 members — **Apr 2025 - Mar 2026**
De Anza College | Led workshops on applied ML, model evaluation, and Python data pipelines

PROJECTS

Accordion | UC Berkeley AI Hackathon 2026 Sponsor Track Winner | Project link — **Jun 2026**
Python, Hugging Face Transformers, SvelteKit

- Built a reversible context-management system for LLM coding agents that visualizes token-weighted context and lets users fold, unfold, or pin sections without deleting stored information.
- Designed keyword, bi-encoder, and cross-encoder relevance ranking; measured 83.3% task completion on SlopCodeBench versus 33.3% for naive compaction at a 100K-token budget.

GazeBoard | Qualcomm x Google LiteRT Edge AI Hackathon | Project link — **Apr - May 2026**
Kotlin, Jetpack Compose, LiteRT/TFLite, ML Kit, CameraX

- Deployed Qualcomm's EyeGaze LiteRT model on a Galaxy S25 Ultra Hexagon NPU, reaching about 8 ms inference latency at 15+ FPS across a 478-landmark face-mesh pipeline.
- Built the offline pipeline from camera capture and face detection through affine gaze calibration, dwell-based tile selection, and text-to-speech output, with zero network permissions.

Clear Dispatch | HackDavis 2026 | Project link — **May 2026**
Python, FastAPI, WebSockets, React, TypeScript, Claude API, ElevenLabs

- Built a real-time 911 dispatcher workflow for call triage, vulnerable-caller flags, voice briefings, and nearest-unit assignment using Haversine routing.
- Implemented a FastAPI/WebSocket backend with event-driven React state and automated Assisted and Surge transitions, while preserving human override at every stage.